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SYSTEMATIC REVIEW

Can I Have My Coffee and Drink It? A Systematic Review and Meta-analysis to Determine Whether Habitual Caffeine Consumption Affects the Ergogenic Effect of Caffeine

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Abstract

Objective The aim was to quantify the proportion of the literature on caffeine supplementation that reports habitual caffeine consumption, and determine the influence of habitual consumption on the acute exercise response to caffeine supplementation, using a systematic review and meta-analytic approach.

Methods Three databases were searched, and articles screened according to inclusion/exclusion criteria. Three-level meta-analyses and meta-regression models were used to investigate the influence of habitual caffeine consumption on caffeine's overall ergogenic effect and within different exercise types (endurance, power, strength), in men and women, and in trained and untrained individuals. Sub-analyses were performed according to the following: acute relative dose (< 3, 3–6, > 6 mg/kg body mass [BM]); whether the acute caffeine dose provided was lower or higher than the mean daily caffeine dose; and the caffeine withdrawal period prior to the intervention (< 24, 24–48, > 48 h).

Results Sixty caffeine studies included sufficient information on habitual consumption to be included in the meta-analysis. A positive overall effect of caffeine was shown in comparison to placebo (standard mean difference [SMD] = 0.25, 95% confidence interval [CI] 0.20–0.30; $p < 0.001$) with no influence of relative habitual caffeine consumption ($p = 0.59$). Subgroup analyses showed a significant ergogenic effect when the caffeine dose was < 3 mg/kg BM (SMD = 0.26, 95% CI 0.12–0.40; $p = 0.003$) and 3–6 mg/kg BM (SMD = 0.26, 95% CI 0.21–0.32; $p < 0.0001$), but not > 6 mg/kg BM (SMD = 0.11, 95% CI − 0.07 to 0.30; $p = 0.23$); when the dose was both higher (SMD = 0.26, 95% CI 0.20–0.31; $p < 0.001$) and lower (SMD = 0.21, 95% CI 0.06–0.36; $p = 0.006$) than the habitual caffeine dose; and when withdrawal was < 24 h, 24–48 h, and > 48 h. Caffeine was effective for endurance, power, and strength exercise, with no influence (all $p \geq 0.23$) of relative habitual caffeine consumption within exercise types. Habitual caffeine consumption did not modify the ergogenic effect of caffeine in male, female, trained or untrained individuals.

Conclusion Habitual caffeine consumption does not appear to influence the acute ergogenic effect of caffeine.

1 Introduction

Caffeine is recognised by the International Olympic Committee (IOC) as one of five dietary supplements with good to strong evidence that it can improve exercise capacity and performance [1]. A substantial body of meta-analytical data shows that acute caffeine intake, at doses of 3–6 mg/kg body mass (BM), exerts an ergogenic effect in exercises performed over a wide range of durations and intensities [2], which explains why approximately 76% of athletes consume it for competition [3]. The primary mechanism by which caffeine may improve exercise performance is through its effect on the central nervous system [4], where caffeine acts as a non-selective antagonist of adenosine A_1 and A_{2A} receptors, increasing the release of norepinephrine and dopamine and intensifying alertness and attention, as well as reducing perceived exertion and pain during exercise [5]. Despite its clear capacity as an ergogenic aid, there appears to be substantial inter-individual variation in the performance

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Key Points

This systematic review and meta-analysis provides evidence that habitual caffeine consumption may not influence the acute ergogenic effect of caffeine across a range of different exercise types (e.g. endurance, strength and power exercises). The same is true for men, women, and trained and untrained individuals.

The ingestion of a pre-exercise caffeine dose lower than the habitually consumed dose was as effective as the consumption of a pre-exercise dose greater than the habitual caffeine dose.

Caffeine supplementation was effective for improving exercise performance only when the pre-exercise caffeine dose was ≤ 6 mg/kg body mass.

Caffeine withdrawal appears unnecessary to benefit from caffeine supplementation.

response following caffeine intake [6, 7]. While most individuals appear likely to experience improvements in exercise performance [8], approximately 33% of participants may not benefit from caffeine supplementation, with some of these even experiencing decreased performance [6].

Habitual caffeine consumption has incited controversy due to contrasting evidence as to whether it influences the acute response to caffeine supplementation. Some studies have shown that habitual caffeine intake blunts some physiological responses, such as increases in plasma epinephrine levels, commonly seen during exercise after acute caffeine supplementation [9, 10]. Thus, it is suggested that chronic caffeine use may reduce the acute benefits of caffeine intake on exercise performance [7, 11]. However, the existing literature on this topic is conflicting, with some studies showing an attenuation of caffeine's acute ergogenic effects when there is also habitual use [12–15] while others do not [16–21]. Some authors have hypothesised that habitual caffeine use may reduce the magnitude of its acute ergogenic effect, but that any reductions in caffeine's ergogenicity may be offset by the ingestion of a pre-exercise dose greater than that habitually consumed [22]. No consensus regarding the influence of habitual caffeine consumption on the acute performance response to its intake currently exists [23].

Conflicting results may be explained by inconsistent thresholds used to categorise the levels of habitual caffeine consumption of participants, and there is a lack of consensus of what should be considered low, moderate, and high-caffeine use [24]. Several previous studies have recruited caffeine naïve participants with little to no habitual caffeine intake to determine caffeine's ergogenic effects [25–28], while a substantial part of the published literature on caffeine supplementation appears not to report habitual intake of participants [29, 30] or reports wide caffeine use (30–850 mg/day) [31]. This makes it difficult to determine whether habitual caffeine intake truly is an important factor that might modify the ergogenic response to caffeine supplementation. Determining the extent to which habitual caffeine consumption may influence the acute performance response to caffeine supplementation can provide important practical information for athletes and researchers alike.

The aim of the current study was to quantify the proportion of the published literature on caffeine supplementation that reports habitual caffeine consumption, and to subsequently determine the influence of habitual caffeine consumption on the acute exercise response to caffeine supplementation using a systematic review and meta-analytic approach.

2 Methods

2.1 Study Eligibility

The study protocol was designed in accordance with Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [32] and the inclusion criteria defined according to PICOS (Population, Intervention, Comparator, Outcomes and Study design) criteria. Only English-language, peer-reviewed, original human studies were included within this review. The study was not preregistered. Initially, the literature was screened to identify all studies investigating the effect of caffeine supplementation on exercise to quantify the proportion of the total evidence base that determined habitual caffeine intake of their sample populations. Data extraction and meta-analysis were subsequently based only on studies that had evaluated habitual caffeine consumption in mg/kg BM/day or that could be calculated as such using participant characteristic data (i.e. provided mean BM and mean absolute daily caffeine consumption). The *population* included healthy human men and women of any age and training status. The *intervention* required acute caffeine supplementation at any dose or in any form (capsule, tablet, beverage, coffee, and gum) prior to an exercise task, with a placebo session or group required as the *comparator.* For the *outcomes,* studies must have evaluated exercise performance or capacity in a randomised single- or double-blind parallel group or cross-over *study design*.

2.2 Search Strategy

An electronic search of the literature was undertaken using three databases (Medline, Embase, and SPORTDiscus) to

identify relevant articles published. The search was initially performed in December 2020, and an updated search was performed at the end of January 2022 to include all indexed articles up to that moment. Studies were searched using the term 'caffeine' AND ('exercise' OR 'performance' OR 'physical performance' OR 'training'). An example search strategy is included in the Electronic Supplementary Material (ESM), Appendix S1. Duplicates were removed before a two-phase search strategy was performed independently by three different researchers (AC, FM, and BG). Phase one assessed the eligibility of the title and abstract of every article generated from the search terms against the inclusion/exclusion criteria. Studies with uncertain suitability were included at this stage and a final decision was reached at the next phase, namely phase two, in which full articles were retrieved and assessed against the eligibility criteria. Reference lists of review articles on this topic were screened to ensure all relevant studies were included. Any differences of opinion relating to study eligibility were resolved through discussion.

2.3 Data Extraction and Variable Categorisation

Data extraction was conducted by AC, FM, GB, and BG using a standardised extraction spreadsheet. Information that was extracted included the following: authors and year of publication, population characteristics (age, BM, sex, training status [determined according to the descriptors used in each study, and subsequently categorised as untrained, trained, or elite], and habitual caffeine consumption), exercise protocol (subsequently classified as endurance [exercise > 30 s in duration], strength [resistance exercises such as bench press or leg press], or power [single or repeated bouts of predominantly anaerobic exercise ≤ 30 s in duration, such as 30-s Wingate] exercise tests), supplementation protocol (dose, delivery method, and ingestion time before exercise), and exercise outcomes (mean and standard deviation [SD]). When studies reported more than one outcome for the same exercise test, a solitary outcome measure was extracted to avoid duplication bias. This was based upon an a priori hierarchy [33]: (1) total work done; (2) mean output throughout the test (i.e. mean power output; mean velocity; mean height); (3) time to completion (performance test)/time to exhaustion (capacity test). The caffeine withdrawal period prior to the intervention was also extracted and categorised (< 24 h, 24–48 h, and > 48 h). All extracted data are available in the ESM (Appendix S2).

2.4 GRADE Certainty of Evidence Classification

Outcomes were rated according to the Grading of Recommendations, Assessment, Development and Evaluations (GRADE) framework [34]. Certainty of evidence could be considered as 'high', 'moderate', 'low', or 'very low' depending on the number of downgrades that were attributed to each of the five topics: (1) risk of bias, (2) imprecision, (3) inconsistency, (4) indirectness, and (5) publication bias. *Risk of bias* was assessed using the revised tool for assessing risk of bias in randomised trials (Cochrane Risk of Bias 2 [ROB 2] tool) [35]. Three reviewers (AC, FM, and BG) independently assessed the risk of bias of each selected study. Disagreements were resolved via discussion and, when necessary, with a fourth reviewer (BS). Risk of bias was judged to be 'low' if all domains were considered low risk, 'some concerns' if at least one domain had 'some concerns' and 'high' if at least one domain was at high risk or more than three domains had 'some concerns' [35]. *Imprecision* was deemed to be present if decision making would be differentially affected when the lower and upper confidence limits (95% confidence intervals [CIs]) were considered as the real effect or if outcomes were calculated from only a few studies with small sample sizes. *Inconsistency* was determined according to heterogeneity measures (I^2 and tau^2). If participants from the included studies differed substantially from the target population (trained individuals), then certainty was downgraded due to *indirectness*. Funnel plots were visually inspected to assess for *publication bias*. Outcomes were upgraded in the presence of either (1) a large magnitude of effect, (2) a dose–response gradient, or (3) an indication that confounding factors would likely reduce rather than increase the magnitude of the effect.

2.5 Statistical Analyses

All analysis were performed using RStudio software (Rstudio version 1.4.1103, PBC, USA). Extracted means and SDs from the caffeine and placebo sessions were transformed into Hedge's *g* standardised mean differences (SMDs) and variances/standard errors (SEs) using the function 'esc_mean_sd' from the 'esc' package. When data were presented in the original studies as SEs, SDs were calculated by multiplying the SE by the square root of the obtained sample size before conversion. An initial three-level random-effects meta-analysis was performed on all exercise outcome data using the 'metagen' function inside the 'meta' package. This model was chosen due to the heterogenous characteristics of the studied populations and exercise tests and due to the large number of outcomes derived from single studies. Outlier detection was performed both manually and via the 'find.outliers' function from the 'metafor' package [36]. Single SMDs were considered outliers when their lower CI was higher than the pooled SMD upper interval, or when one study's upper CI was lower than the pooled SMD lower interval [36].

Further three-level meta-analyses were performed on outcomes within endurance, strength, or power exercise tests, and

within studies including males or females (except for those which combined both sexes) and trained or untrained individuals. Due to the small number of studies including individuals classified as elite ($n=3$), these were included within the trained group. Meta regressions were then performed within each of the eight main meta-analyses (overall, endurance, strength, power, males, females, trained, untrained) with relative habitual caffeine consumption (in mg/kg BM/day) as a continuous variable. The overall influence of (1) acute relative caffeine dose (three levels: < 3 mg/kg BM, 3–6 mg/kg BM, and > 6 mg/kg BM), (2) whether the acute caffeine dose provided was lower or higher than the mean daily dose of caffeine habitually ingested by participants, and (3) the caffeine withdrawal period prior to the intervention (< 24 h, 24–48 h, and > 48 h) was also evaluated through meta regressions with these factors as moderators with the 'rma' package of the 'metafor' package. SMDs of < 0.2, 0.2–0.5, 0.5–0.8, and > 0.8 were classified as very small, small, medium, and large effects [37]. Heterogeneity was assessed using the I^2 statistic and is reported alongside tau^2 values. Values of ≤ 50% indicate low heterogeneity, 50–75% moderate heterogeneity, and > 75% high heterogeneity. Statistical significance was set at $p<0.05$.

3 Results

3.1 Study Search and Characteristics

The primary search generated 8037 results (Fig. 1). Following removal of duplicates ($n=2238$), phase one resulted in the exclusion of 5484 records. The remaining papers ($n=315$) were screened in their entirety for suitability. A total of 246 met the initial inclusion criteria, of which 186 (~ 76%) did not report any information about habitual caffeine consumption of the volunteers. Only 61 studies included sufficient information on habitual consumption and were taken forward to the meta-analysis. However, one study was excluded due to not providing means and standard deviations [38] and another due to a large effect size that contributed to high heterogeneity [25]. The total number of studies (n) included was 59 [13, 14, 16–20, 25, 26, 28, 39–87], with a total of 198 outcome measures (k). Data from a total of 1137 individuals were included in this meta-analysis. Of these, 958 were men and 179 were women; 718 were trained, 400 were untrained, and 19 were classified as elite.

3.2 Meta-analysis

A small positive effect of caffeine was shown in comparison to placebo (SMD = 0.25, 95% CI 0.20–0.30; $n=59$, $k=198$; $p<0.001$; $I^2=0.0\%$; Fig. 2). Due to the large number of studies and outcomes, visual presentation of the overall meta-analysis result is as a funnel plot and not a forest plot. Meta-regression showed no influence ($p=0.59$) of relative habitual caffeine consumption (in mg/kg BM/day) on the effects of caffeine ($R^2=0\%$, estimate = − 0.01, 95% CI − 0.04 to 0.02). Subanalysis by dose (Fig. 3E) showed that there was a significant effect of caffeine supplementation when doses were < 3 mg/kg BM (SMD = 0.26, 95% CI 0.12–0.40; $k=21$; $p=0.003$; $I^2=0.0\%$), between 3 and 6 mg/kg BM (SMD = 0.26, 95% CI 0.21–0.32; $k=164$; $p<0.0001$; $I^2=0.0\%$), but not > 6 mg/kg BM (SMD = 0.11, 95% CI − 0.07 to 0.30; $k=13$; $p=0.23$; $I^2=0.0\%$). The meta-regression showed no influence of dose on the effects of caffeine ($R^2=0\%$; $p=0.34$). There was a significant effect of caffeine both when the acute dose was higher (SMD = 0.26, 95% CI 0.20–0.31; $k=171$; $p<0.001$; $I^2=0.0\%$; Fig. 3D) and lower (SMD = 0.21, 95% CI 0.06–0.36; $k=27$; $p=0.006$; $I^2=0.0\%$; Fig. 3D) than the habitually consumed daily dose of caffeine, with meta-regressions showing no influence ($R^2=0\%$; $p=0.55$). Analyses performed considering the time of caffeine withdrawal prior to the acute caffeine interventions (Fig. 3F) showed significant SMDs when withdrawal was < 24 h (SMD = 0.25, 95% CI 0.16–0.33; $k=74$; $p<0.001$; $I^2=0.0\%$), between 24 and 48 h (SMD = 0.26, 95% CI 0.19–0.33; $k=114$; $p<0.001$; $I^2=0.0\%$), and > 48 h (SMD = 0.22, 95% CI 0.04–0.39; $k=6$; $p=0.01$; $I^2=0.0\%$), with no effect of this moderator shown by the regression model ($R^2=0\%$; $p=0.89$).

Meta-analyses were performed for endurance ($n=23$, $k=44$), power ($n=23$, $k=76$), and strength exercises ($n=17$, $k=78$) individually. Caffeine was effective for all exercise types (Fig. 3A), namely endurance (SMD = 0.25, 95% CI 0.15–0.35; $p<0.0001$; $I^2=0.0\%$), power (SMD = 0.25, 95% CI 0.17–0.34; $p<0.0001$; $I^2=0.0\%$), and strength (SMD = 0.25, 95% CI 0.15–0.34; $p<0.0001$; $I^2=0.0\%$) exercise. Meta-regressions within these exercise type subgroups showed no influence (all $p\geq0.23$) of relative habitual caffeine consumption on the effects of caffeine (endurance: $R^2=0\%$, estimate = 0.03, 95% CI − 0.03 to 0.08; power: $R^2=0\%$, estimate = − 0.04, 95% CI − 0.11 to 0.03; strength: $R^2=0\%$, estimate = − 0.02, 95% CI − 0.06 to 0.02).

Caffeine was also effective when considering only male (SMD = 0.26, 95% CI 0.20–0.32; $n=44$, $k=156$; $p<0.0001$; $I^2=0.0\%$; Fig. 3C) or female (SMD = 0.23, 95% CI 0.10–0.35; $n=9$, $k=36$; $p=0.0005$; $I^2=0.0\%$; Fig. 3C) participants, and for trained (SMD = 0.27, 95% CI 0.21–0.33; $n=37$, $k=160$; $p<0.0001$; $I^2=0.0\%$; Fig. 3B) and untrained (SMD = 0.18, 95% CI 0.08–0.29; $n=17$, $k=35$; $p=0.0006$; $I^2=0.0\%$; Fig. 3B) individuals. Regressions showed no influence of habitual caffeine consumption (all $R^2=0\%$; all $p\geq0.61$) within any of these subgroups.

3.3 Risk of Bias

None of the outcomes here were classified as having a low risk of bias, while 82% were considered as having some

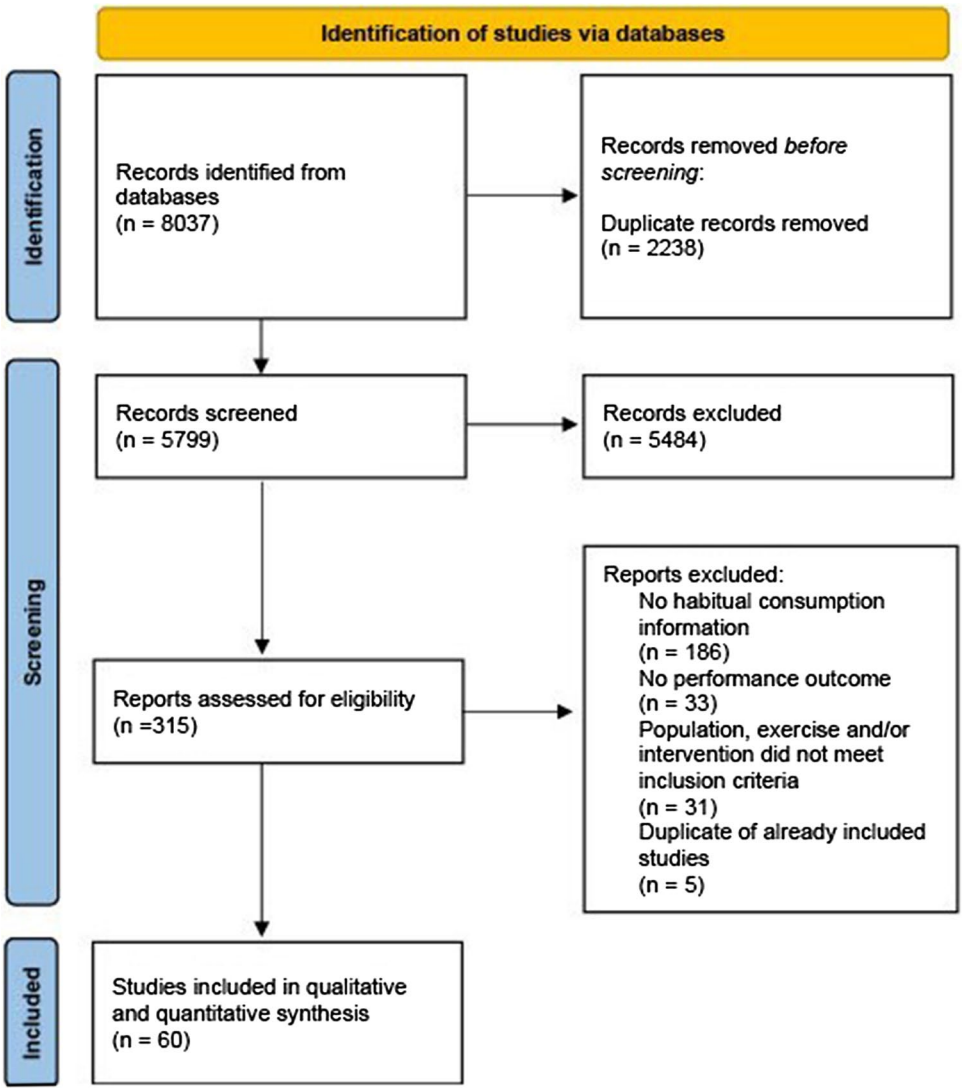

Fig. 1 Flowchart of the search strategy and study selection

concerns and the remaining 18% as having a high risk of bias. Most of the included studies had issues (some concerns or high risk) in the first (57.4%) and fifth (100%) domains, due to insufficient reporting of randomisation method and lack of pre-registration. Most of the studies were judged as having a low risk of bias arising from the second (98.4%), third (75.4%), and fourth domains (77%) of the ROB 2 tool. A summary of these results is presented in Fig. 4.

3.4 GRADE Certainty of Evidence Classification

Most outcomes were considered to have a 'moderate' certainty of evidence (11 out of 16), while the remaining ones were classified as 'high' (see the ESM, Appendix S3). No substantial asymmetry was detected in the funnel plots (see the ESM, Appendix S4) after visual examination. Therefore, it was assumed that no publication bias or small study effects existed. Imprecision was deemed to have occurred in four out of the 16 outcomes with moderate certainty, mostly due to the small number of studies included or large CIs. The other seven were considered to contain indirectness, which was the case when more than 25% of the included outcomes were derived from studies with non-specifically trained individuals. Evidence for strength and power exercise outcomes and trained, untrained, and male individuals was considered as providing high certainty.

4 Discussion

The results of this systematic review and meta-analysis showed a small positive overall effect of caffeine supplementation on exercise outcomes (endurance, power, and strength) with no influence of habitual caffeine consumption. There was also no influence of habitual caffeine consumption on the ergogenic effect of caffeine for male or

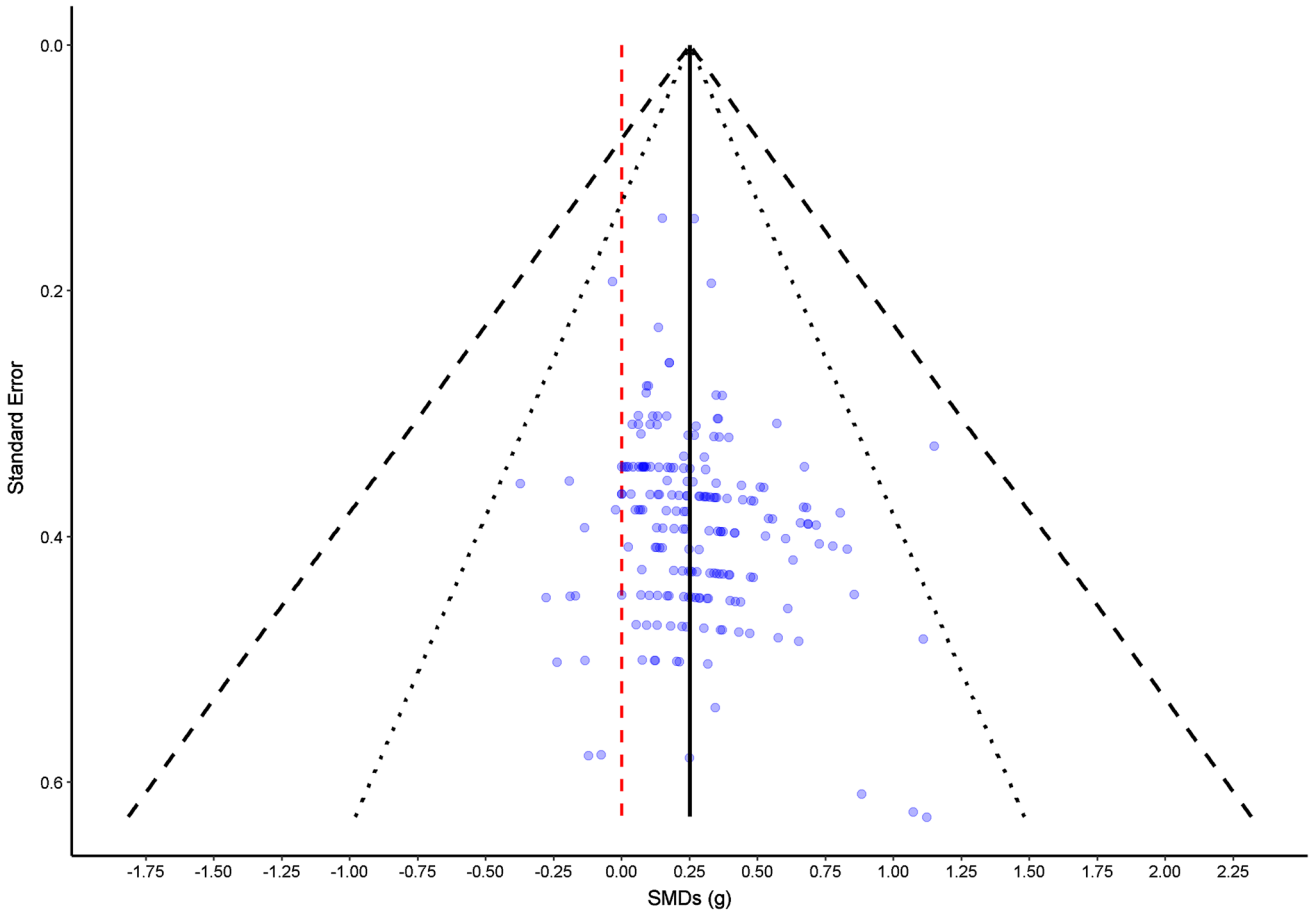

Fig. 2 Funnel plot illustrating standardised mean difference (SMD) effect sizes for exercise outcomes relative to within-study standard errors. Each circle represents one study; darker shading of circles indicates multiple studies. The solid black line represents the mean SMD; the dashed red line indicates an SMD of zero. The inner and outer dashed lines represent 95% and 99% confidence intervals

female participants, or for trained or untrained individuals. A positive effect was shown when the acute caffeine dose was ≤ 6 mg/kg BM, but not > 6 mg/kg BM. Additionally, caffeine appears ergogenic regardless of whether the acute caffeine dose is lower or higher than the daily dose of caffeine, and of the caffeine withdrawal period. This systematic literature search showed that only 24% (60 of 246 studies) of caffeine and exercise studies reported the mean habitual caffeine consumption of their volunteers.

4.1 Habitual Caffeine Consumption

Only a small proportion of the caffeine literature determined the habitual caffeine consumption of participants, and even fewer directly investigated its influence on the acute ergogenic effect of caffeine supplementation [12, 14, 16–21]. This was the first study to investigate, using a systematic review and meta-analytic approach, the influence of habitual caffeine intake on exercise performance and capacity following acute caffeine supplementation. The data showed that relative habitual caffeine consumption does not affect the acute ergogenic effect of caffeine supplementation, regardless of the biological sex and training status of the individuals. The lack of influence of habitual caffeine intake on the exercise response to caffeine supplementation agrees with several studies on this topic, which show that individuals with different levels of daily caffeine consumption experience similar benefits from acute caffeine intake, regardless of their daily level of caffeine intake [16–21]. This information is important for athletes who regularly consume caffeine, as it suggests they may continue to do so without significant impact to the acute ergogenic effects of their pre-training or pre-competition caffeine dose. It must be acknowledged that contrasting evidence exists, with some studies showing that chronic daily caffeine intake may reduce the acute ergogenic effect of caffeine supplementation [13, 15]. In these studies, participants appeared to show a certain level of tolerance when consuming 3 mg/kg BM for 20–28 days, which reduced, but did not entirely eliminate, the ergogenic effects of an acute dose of caffeine (3 mg/kg BM) in individuals

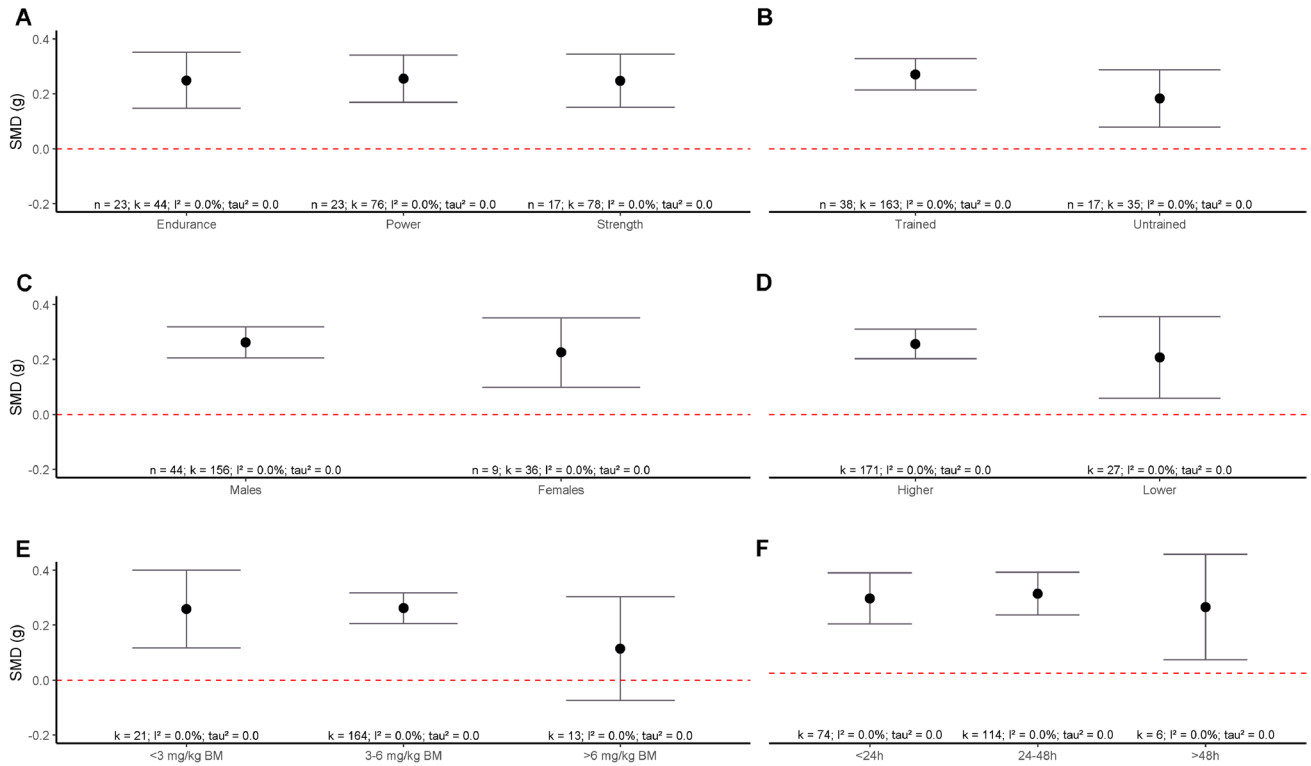

Fig. 3 Plots of standardised mean differences (SMDs) and 95% confidence intervals according to **A** exercise type, **B** training status, **C** biological sex, **D** whether the acute caffeine dose was higher or lower than the habitual caffeine consumption, **E** acute caffeine dose, and **F** duration of caffeine withdrawal. I^2 percentage of the heterogeneity that is due to differences between studies, *k* number of included outcomes, *n* number of included studies, tau^2 heterogeneity between studies

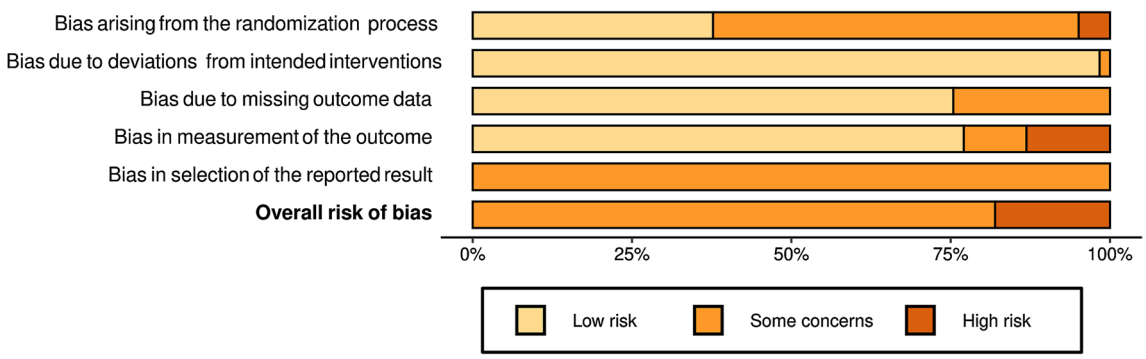

Fig. 4 Risk of bias presented as percentages across all included studies for the five main domains of evaluation (figure was created using robvis and is in a colour-blind-friendly colour scheme)

who were previously caffeine naïve or low consumers [13, 15]. Despite showing that daily caffeine consumption may induce a progressive tolerance to the ergogenic effect of its acute intake [13, 15], it is important to recognise that these short-term chronic intervention studies may not reflect the true habitual caffeine consumption of individuals and, consequently, should not be extrapolated to what is seen in individuals with long-term chronic caffeine use. Although the results here suggest that habitual caffeine consumption does not influence the acute performance response to its ingestion at a group level, it is possible that there may be some variation in this response at the individual level. Results should not necessarily be extrapolated to all individuals, and athletes are advised to test caffeine for themselves throughout training to determine their own individual responsiveness.

It has been suggested that habitual caffeine use may reduce the magnitude of its acute ergogenic effect, but that any reductions in caffeine's ergogenicity may be offset by the

ingestion of a pre-exercise dose greater than that habitually consumed [22]. This appears logical as, according to some animal model studies [38, 39], the most likely mechanism through which habituation to the ergogenic effects of caffeine could develop includes an increase in the amount of adenosine receptors in the central nervous system. Thus, more caffeine would be required to reach a similar level of antagonism. The meta-analytical data here do not support this hypothesis, with performance gains irrespective of whether the acute caffeine dose was lower or higher than that habitually consumed. In fact, individuals looking to consume more caffeine than their habitual dose to obtain ergogenic benefits should be aware that caffeine benefited exercise performance only when consumed up to 6 mg/kg BM, but not > 6 mg/kg BM. This is in line with experimental studies that have shown significant positive effects of caffeine with doses of 3 and 6 mg/kg BM, but not 9 mg/kg BM [40]. A possible explanation for this lack of effect at doses > 6 mg/kg BM is that the intake of high doses of caffeine may increase the risk of side effects, such as nausea, anxiety, insomnia, and restlessness [1]. Some caution is needed when interpreting the results of these analyses here as they were conducted using group mean values of daily habitual caffeine consumption and therefore most individuals will have fallen above or below these means. Furthermore, only five studies with doses > 6 mg/kg BM were included. Experimental studies specifically designed to test the hypothesis that acute caffeine doses exceeding an individual's habitual dose positively affect exercise performance should be conducted. Collectively, the findings of this study suggest that habitual caffeine consumption may not be as important as previously suggested and that athletes should not exaggerate their pre-exercise caffeine dose to elicit ergogenic effects.

4.2 Exercise Type, Biological Sex and Training Status

The small ergogenic effects of caffeine were similar between exercise types, namely endurance, power, and strength exercises, which is in line with previous meta-analytical evidence indicating that supplementation benefits a wide range of exercise types [2, 41–44]. Importantly, meta-regressions within these exercise types confirmed the absence of any moderating effect of habitual caffeine intake on the acute ergogenic response. This is in agreement with experimental studies that showed individuals with different levels of daily caffeine consumption benefited equally from caffeine supplementation in endurance [17], power [18], and strength [21] exercise tests. Similarly, habitual caffeine consumption did not modify the ergogenic effect of caffeine whether the participants were male or female, or trained or untrained. The response to caffeine supplementation may potentially be influenced by several factors other than its habitual use, such as genetics, expectancy, biological sex, training status, age, caffeine dose, supplementation timing, and caffeine delivery method [7, 23]. Therefore, a single factor, such as habitual caffeine use, is unlikely to have a major influence on the effect of caffeine supplementation.

4.3 Caffeine Withdrawal

The belief that habitual caffeine intake influences the performance benefits following its supplementation has led to recommendations to abstain from caffeine consumption for the days before competitions to maximise its ergogenic effect [45]. Nevertheless, these data are in direct disagreement with this notion and suggest that caffeine withdrawal is unnecessary to elicit performance improvements with caffeine supplementation, with significant improvements when withdrawal was < 24 h, between 24 and 48 h, and > 48 h. This agrees with experimental data showing that up to 4 days of caffeine withdrawal does not modify the acute benefits of its supplementation on exercise in habitual caffeine users [46, 47]. Thus, what seems apparent is that even with little to no withdrawal, there is an ergogenic effect of caffeine. As such, these data suggest that pre-competition caffeine withdrawal is unnecessary to elicit a performance-enhancing effect from pre-exercise caffeine supplementation. In addition, it should be noted that caffeine withdrawal can cause side effects among habitual users, such as severe headaches, fatigue, lethargy [46], drowsiness, impaired concentration, depressed mood, anxiety, and irritability [5]. Experiencing these symptoms in the days leading up to a competition can impair the quality of the training and even negatively affect mental factors such as confidence, and should be avoided at all costs.

4.4 Limitations

One of the main limitations of this meta-analysis is that it was conducted using group means of both BM and habitual caffeine intake, meaning most participants will have fallen above or below these group means. Where possible we included groups that had already been separated according to habitual caffeine consumption [17, 19], although these amounted to very few studies. Indeed, another important limitation is that very few studies specifically set out with the aim of determining the effect of habitual consumption on exercise outcomes, hence a lack of detailed reporting. We urge all future work with caffeine supplementation to determine and report participants' habitual caffeine consumption in as much detail as possible, including individual data. Similarly, it is important to recognise that we used group means for exercise performance, and there is evidence of inter-individual variation in the ergogenic effect of caffeine supplementation [6, 7]. Very few studies ($n = 9$) that met the

inclusion criteria were conducted with female participants, which means that these results should be extrapolated to women with caution. Thus, further studies are needed to investigate the influence of habitual caffeine consumption on caffeine's ergogenic effect in women. Habitual caffeine consumption is usually determined via Food Frequency Questionnaires and standard caffeine quantities based upon food frameworks and available references. However, there is large variability in the caffeine content within and between caffeine sources [48, 49]. Therefore, it is possible that the habitual caffeine intakes of the participants in these studies are not entirely accurate [50]. However, these 'best estimates' can still be useful, particularly if combined with CIs. Finally, none of the included studies were classified as having a low risk of bias, and this can be attributed to insufficient reporting of randomisation method and lack of pre-registration. It should be highlighted that these are not currently common practices in the field of sports science. In view of this, we recommend that future studies in this area report the randomisation method and perform pre-registration.

4.5 Practical Implications

The current data can be used to better guide athletes' and coaches' practices regarding caffeine supplementation to improve exercise performance. Current evidence suggests that caffeine improves exercise performance and that athletes should preferably consume a caffeine dose ≤ 6 mg/kg BM, prior to competition or important training sessions. It would be wise to be cautious about ingesting high doses of caffeine (> 6 mg/kg BM) before exercise, as this does not appear to have an additional ergogenic effect and may increase the risk of undesired side effects. Caffeine supplementation is expected to improve athletic performance in endurance (e.g. running, cycling, and triathlon), strength (e.g. weightlifting and powerlifting), and power (e.g. team and combat sports) sports, so it seems to be useful in almost all exercise modalities. Likewise, both male and female athletes can benefit from caffeine supplementation without worrying about their habitual caffeine consumption, while even untrained individuals can expect an ergogenic effect. Considering the evidence that habitual caffeine consumption does not influence its ergogenic effect, both low and high habitual caffeine users can equally benefit from pre-competition caffeine supplementation, without the need to reduce their daily intake of caffeine sources or increase the pre-exercise caffeine dose to achieve an improved performance. Therefore, caffeine supplementation can be used before training sessions aiming to enhance training quality (e.g. increase training volume or intensity) and potentially some physiological and performance adaptations to exercise, without any harm to its pre-competition use. Despite being traditionally used by athletes and supported by coaches, pre-competition caffeine withdrawal does not enhance caffeine's ergogenic effect as commonly thought and appears to be an unnecessary practice. Thus, these data suggest that there is no need to implement caffeine withdrawal in research or practice.

5 Conclusion

This systematic review and meta-analysis suggest that habitual caffeine consumption does not influence the ergogenic effect of caffeine supplementation across a range of different exercise types, or for male, female, trained, and untrained individuals. The ingestion of a pre-exercise caffeine dose lower than the habitually consumed dose seems to be equally effective as the consumption of a pre-exercise dose greater than the habitual caffeine dose. Caffeine withdrawal does not enhance the acute ergogenic effect following caffeine intake and appears unnecessary. Lastly, pre-exercise caffeine doses < 3 mg/kg BM and between 3 and 6 mg/kg BM were effective in improving exercise performance, whereas a caffeine dose > 6 mg/kg was not. These data should be used by practitioners to guide individual practices regarding caffeine supplementation as well as by researchers to guide further research in the field [13, 14, 16–20, 25, 26, 28, 51–93].

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Declarations

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Conflict of interest Several of the authors (GB, BS) have previously received caffeine supplements at no cost from a national supplement company (Farmácia Analítica, Rio de Janeiro, Brazil) for work unrelated to the current article. Farmácia Analítica have not had any input (financial, intellectual, or otherwise) into this review. The remaining authors report no conflict of interest.

Author contributions BS is responsible for the conception of the work. FM performed the searches. AC, FM, BG, and GB performed the screening, and AC, FM, and BG performed the data extraction. GB performed the data analysis. AC and BS are responsible for the initial writing of the manuscript, and all authors were involved in the editing process. All authors approved the final version of the manuscript.

Data availability Extracted data are available in a supplementary file, and analysis codes are available upon request.

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